

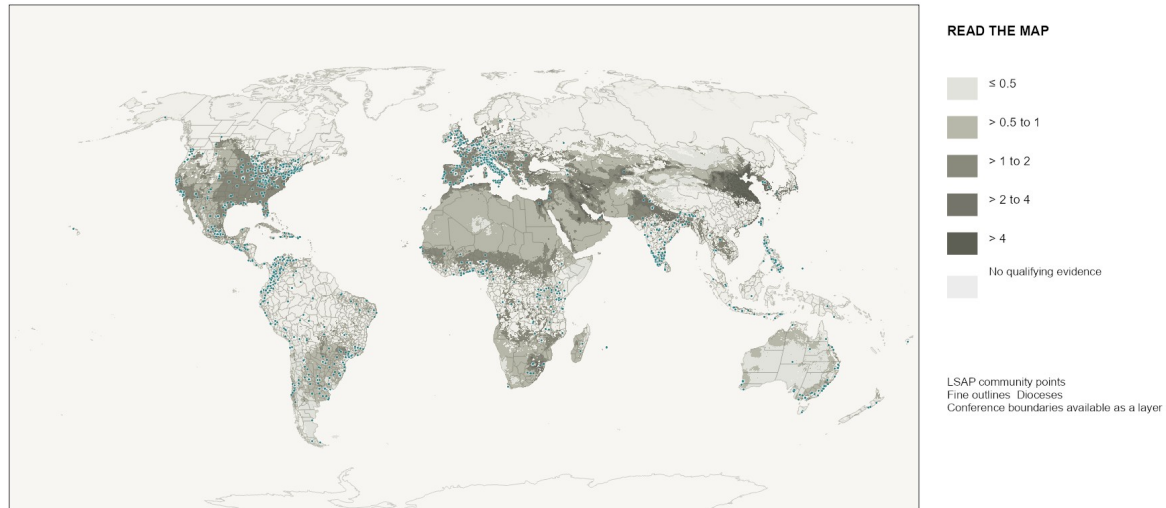
Annual NO2 atmospheric composite

This report explains the source data, executed workflow and interpretation of annual no2 atmospheric composite. It documents the existing ArcGIS Pro output, including its coverage limits and retained ecclesiastical context. (Glissenaar et al., 2024, 2025; KNMI, n.d.).

Map P12 | Documentation date 6 October 2026 | Units 10^{15} molecules/cm² | Air year 2021

Annual NO2 atmospheric composite

Annual 2021 · 10^{15} molecules/cm²



Native geographic grid preserved; no-data cells are transparent. Diocesan statistics use cell areas and fractional overlaps.

KNMI ESA CCI+ TROPOMI L3 v1.26

LSAP POLLUTION ATLAS / COPIED ECCLESIASTICAL AND MEMBER FOUNDATIONS / P12

Figure P12 World overview exported from the completed ArcGIS Pro layout.

Native composite surface; only cells passing temporal and retrieval filters are shown.

All analyses use copies of 2,874 dioceses and 2,739 LSAP member points. Copies of 122 episcopal conference features and 194 country features preserve ecclesiastical context. Stable DIO_ID links outputs to the copied boundaries. Original foundational data were not edited (LSAP Pollution Atlas, 2026, foundation_manifest.json).

Analysis workflow

The following numbered sequence describes the executed ArcPy and Python analyses as a typed ModelBuilder style workflow. Custom raster, station aggregation and geometry operations are script steps, not a claim that an ArcGIS ModelBuilder model file was run. Source data: (Glissenaar et al., 2024, 2025; KNMI, n.d.).

1. Download all twelve 2021 KNMI ESA CCI+ TROPOMI L3 v1.26 monthly files (fv1.0 filename version), native 900 × 1800 grid.
2. Select monthly cells with qa_L3=1, land/transition mask, finite column value and fractional effective monthly coverage × calendar days ≥10. This effective coverage is not a literal count of valid days.
3. Calculate calendar-day-weighted means of qualifying monthly columns. Require at least nine qualifying months and representation from all four quarters. Preserve negative retrieval values; divide by 10¹⁵ to express molecules/cm² in readable units.
4. Export annual concentration and qualifying-month-count rasters; audit both CRS and grids. Aggregate annual concentration using geodesic cell areas and fractional overlaps. Retain weighted spatial median and full-polygon valid coverage.
5. Display NO2_Annual_2021.tif with breaks at 0.5, 1, 2 and 4 ×10¹⁵ molecules/cm² above the country backdrop. Overlay copied LSAP points, diocesan outlines and optional conferences. Companion diocesan statistics are in P02.

Area weighting and coverage

Both concentration rasters use geographic cells, not equal area cells. For diocesan aggregation: weighted mean = $\Sigma(c_i \times a_i) / \Sigma(a_i)$, summing only valid cells. a_i is geodesic row cell area multiplied by native grid overlap fraction. This treats concentration as constant within a cell; geographic boundary fractions approximate true geodesic overlap within boundary cells. Spatial median uses the same area weights.

Coverage uses the full polygon denominator, including the PM25 extent correction at 60°S–70°N. AirDisplay requires at least 50 percent valid coverage while all computed statistics remain in attributes. The NO2 implementation uses global native grid polygon overlap for its denominator; PM25 partial extent correction uses the ratio of clipped to full geodesic polygon area. These are documented numerical approximations, not population weighting.

Output fields and files

Field or band	Meaning
Band 1	Annual qualifying monthly composite at native 0.2 degree grid
NO2_ValidMonths_2021.tif	Separate qualifying month count raster retained

Analysis dataset: Rasters/NO2_Annual_2021.tif. Styles/P12.lyrx and Styles/P12.mapx preserve layer and map definitions. Map_Exports/P12_World_overview.png preserves the layout preview. The editable project is LSAP_Pollution.aprx.

Interpretation and data handling

A cell with nine qualifying months is an observed-month estimate, not complete annual sampling. Tropospheric column density is different from surface concentration or personal exposure. KNMI source.

Cartographic classes are descriptive display intervals, not regulatory or ecological quality grades. The pollution ramp is #E1E2DB, #B7B8A9, #898A7C, #74746A and #5D5E54. Lower oxygen and shorter proximity distances use the dark end. Foundation member dots use teal #247C80; proximity maps color member points by distance.

References

Glissenaar, I. A., Anglou, I., Boersma, K. F., & Eskes, H. (2024). ESA CCI+ TROPOMI L3 monthly mean NO2 columns [Data set]. Royal Netherlands Meteorological Institute. Version 1.26, filenames fv1.0; 2021 monthly files. [Source link](#)

Glissenaar, I., Boersma, K. F., Anglou, I., Rijdsdijk, P., Verhoelst, T., Compernelle, S., Pinardi, G., Lambert, J.-C., Van Roozendaal, M., & Eskes, H. (2025). TROPOMI Level 3 tropospheric NO2 dataset with advanced uncertainty analysis from the ESA CCI+ ECV precursor project. *Earth System Science Data*, 17, 4627–4650. [Source link](#)

KNMI. (n.d.). ESA CCI+ NO2 TROPOMI level 3 data. Download and version documentation. Accessed 6 October 2026. [Source link](#)

Esri. (n.d.). Calculate Geometry Attributes. ArcGIS Pro documentation. Reference for geodesic area measurement. Accessed 6 October 2026. [Source link](#)

LSAP Pollution Atlas. (2026). Internal source copies, executed analysis scripts and validation records [Project records]. LSAP_Pollution.aprx; FoundationCopies.gdb; PollutionAnalysis.gdb; foundation_manifest.json; raster_audit.json; freshwater_aggregate_QA.json; waste_QA.json; cleanup_audit.json; project_QA.json. Project root: Atlas Analyses/Pollution. Internal provenance is distinct from published source citations.